

# Multi-Agent Scientific Paper Review

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|                   |                                                                                      |
|-------------------|--------------------------------------------------------------------------------------|
| File              | PIIS0022391325009370.pdf                                                             |
| Words             | 2,033                                                                                |
| Sections Detected | preamble, methodology, abstract, discussion, results, conclusions, references, title |
| Review Mode       | FAST                                                                                 |
| Model             | qwen3:8b                                                                             |

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## Editorial Decision

REJECT

Weighted Score: 2.3 / 5.0  
Confidence: 72%

## Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

| Criterion                | Score | Rating    | Visual |
|--------------------------|-------|-----------|--------|
| Originality & Novelty    | 3.8   | Good      | ■■■■■  |
| Manuscript Structure     | 2.0   | Weak      | ■■■■■  |
| Methodology & Equations  | 2.0   | Weak      | ■■■■■  |
| Statistical Analysis     | —     | Skipped   | —      |
| Results & Evidence       | 2.0   | Weak      | ■■■■■  |
| Discussion & Conclusions | 1.5   | Very Weak | ■■■■■  |
| Figures & Tables         | 3.5   | Good      | ■■■■■  |
| Scientific Writing       | 2.5   | Weak      | ■■■■■  |
| References & Citations   | —     | Skipped   | —      |
| Ethics & Transparency    | 1.5   | Very Weak | ■■■■■  |

# Detailed Agent Reviews

## TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 3.8 / 5.0 | Confidence: 65%

**Strengths:**

- [TITLE] Title detected and available for review.
- [ABSTRACT] Abstract detected and available for review.

**Weaknesses:**

- [KEYWORDS] Keywords were not detected or fewer than three were found.

**Minor Comments:**

- [KEYWORDS] Keywords were not detected or fewer than three were found.

**Recommendations:**

- [KEYWORDS] Provide 4-8 indexing terms after the abstract.

## StructureReviewer

Manuscript structure and logical coherence

Score: 2.0 / 5.0 | Confidence: 90%

**Strengths:**

- Clear title and abstract
- Well-defined methodology section

**Weaknesses:**

- Missing introduction and literature review sections
- Poor logical flow between sections
- Inconsistent section balance
- Lack of clear research problem and hypothesis

**Major Comments:**

- The manuscript lacks a critical introduction and literature review section, which are essential for establishing context and research problem.
- The logical flow of the manuscript is severely disrupted due to missing sections and poor organization.
- The research problem and hypothesis are not clearly stated or declared.
- The objectives are not explicitly stated and do not align with the conclusions.

**Minor Comments:**

- Section balance is uneven, with some sections disproportionately long or absent.
- The references section is incomplete and lacks proper formatting.

**Recommendations:**

- Add an introduction and literature review section to provide context and establish the research problem.
- Reorganize the manuscript to follow the standard structure: Introduction → Literature → Methods → Results → Discussion → Conclusions.
- Clearly state the research hypothesis and objectives in the introduction.
- Ensure that the objectives are explicitly addressed in the conclusions.
- Complete and properly format the references section.

## MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 2.0 / 5.0 | Confidence: 50%

**Strengths:**

- Clear integration of digital techniques for prosthetic and implant planning
- Use of CBCT and CAD-CAM for precision in dental applications

**Weaknesses:**

- Lack of detailed methodology and experimental setup
- No equations or mathematical formulations provided
- Insufficient baseline comparison and bias analysis

**Major Comments:**

- The methodology is not clearly described, making it difficult to replicate the work.
- The experimental setup and validation are not sufficiently detailed.

**Minor Comments:**

- The problem statement lacks specific objectives and challenges.
- The model and method sections are too vague.

**Recommendations:**

- Provide a detailed description of the workflow, including specific steps and tools used.
- Include equations or mathematical formulations if applicable.
- Clarify the experimental setup, dataset, and validation process.

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**StatisticsReviewer**

*Statistical methods, tests, and reporting quality*

■ Skipped: No statistical content detected. Agent skipped — does not penalize score.

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**FiguresTablesEquationsReviewer**

*Figures, tables, equations — quality and coherence with text*

Score: **3.5 / 5.0** | Confidence: 62%

**Strengths:**

- 2 figure references were detected in the manuscript.

**Weaknesses:**

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

**Minor Comments:**

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

**Recommendations:**

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
- 

**ResultsReviewer**

*Results validity and empirical support*

Score: **2.0 / 5.0** | Confidence: 80%

**Strengths:**

- Mentions of hypervolume/spread indicators suggest quantitative analysis

**Weaknesses:**

- Results are not clearly tied to figures/tables
- No detailed quantitative metrics provided
- No comparison with baseline or state-of-the-art methods

**Major Comments:**

- The results section lacks sufficient detail to support the stated objectives. Quantitative results are mentioned but not sufficiently elaborated or tied to figures/tables.
- No baseline or state-of-the-art comparison is provided, which is critical for evaluating the novelty and effectiveness of the proposed workflow.

**Minor Comments:**

- The phrase 'workflow' is used ambiguously in the context of patient selection criteria.

**Recommendations:**

- Include detailed quantitative metrics such as MSE, RMSE, CV, or hypervolume values with corresponding figures/tables.
- Provide a clear comparison with existing methods or baselines to contextualize the results.
- Clarify the criteria for patient selection and the role of 'workflow' in the context of the study.

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## DiscussionConclusionsReviewer

*Discussion depth and conclusions validity*

Score: **1.5 / 5.0** | Confidence: 80%

**Strengths:**

- Partial references to relevant literature

**Weaknesses:**

- Lack of detailed discussion on results in context
- No clear limitations stated
- Conclusions not directly supported by results
- No specific future work directions

**Major Comments:**

- The Discussion section lacks a comprehensive comparison with prior work and does not interpret results in the context of existing literature.
- The Conclusions section does not directly derive from the results and fails to answer the original hypothesis.
- No clear statement of the study's limitations or contributions to the field.

**Minor Comments:**

- The Discussion section is incomplete and lacks depth.
- The Conclusions section is vague and not quantified.

**Recommendations:**

- Expand the Discussion to include a detailed comparison with prior studies and interpret results in the context of existing literature.
- Revise the Conclusions to ensure they are directly derived from the results and clearly answer the research hypothesis.
- Clearly state the study's limitations and contributions to the field.

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## WritingReviewer

*Scientific writing quality, clarity, and style*

Score: **2.5 / 5.0** | Confidence: 70%

**Strengths:**

- Clear technical terms defined on first use (e.g., CAD/CAM)
- Use of academic tone in some sections

**Weaknesses:**

- Overuse of passive voice
- Unexplained acronyms

- Vague quantifiers
- Incomplete sentences
- Poor paragraph structure

**Major Comments:**

- The manuscript contains multiple instances of passive voice that obscure the clarity of the methodology.
- Acronyms like CAD/CAM are used without definition, which is a major issue for clarity.
- The text is incomplete and lacks coherence, with many run-on sentences and missing sections.

**Minor Comments:**

- Some technical terms are defined on first use, which is a positive aspect.
- The abstract is cut off and incomplete, which affects the overall readability.

**Recommendations:**

- Define all acronyms on first use.
- Revise passive voice constructions to improve clarity.
- Complete the manuscript text and ensure all sections are properly structured.
- Ensure all sentences are complete and grammatically correct.
- Use active voice where possible to enhance clarity and flow.

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## ReferencesReviewer

*References quality, recency, and relevance*

Score: **0.0 / 5.0** | Confidence: 0%

**Weaknesses:**

- The references section is completely missing, which severely limits the manuscript's credibility and reproducibility.

**Major Comments:**

- The absence of a references section is a critical flaw that prevents evaluation of the manuscript's scholarly rigor and contribution to the field.

**Recommendations:**

- The authors are strongly advised to include a complete references section with properly cited sources to meet the standards of Q1 journals.

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## EthicsLimitationsReviewer

*Research ethics, limitations, and transparency*

Score: **1.5 / 5.0** | Confidence: 80%

**Strengths:**

- The manuscript presents a novel digital workflow for prosthetically guided implant positioning, which is a significant contribution to the field.

**Weaknesses:**

- The manuscript lacks comprehensive reporting on ethical considerations, which is essential for transparency and reproducibility.
- Limitations are not explicitly stated or discussed, which undermines the validity of the conclusions.
- The manuscript does not address reproducibility, such as data and code availability, which is critical for scientific rigor.

**Major Comments:**

- The manuscript fails to report ethical approval, data availability, conflict of interest, funding sources, and author contributions, which are essential for transparency and reproducibility.
- Limitations are not discussed in detail, which is a major omission in the discussion section.

**Minor Comments:**

- The manuscript could benefit from a more detailed description of the methodology and results.

**Recommendations:**

- Include ethical approval details for any human or animal studies.
- Provide a data availability statement and disclose any potential conflicts of interest.
- Clearly state and discuss the limitations of the study in the discussion section.
- Address reproducibility by providing access to data and code.

## Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] The manuscript lacks a critical introduction and literature review section, which are essential for establishing context and research problem.
- [2] The logical flow of the manuscript is severely disrupted due to missing sections and poor organization.
- [3] The research problem and hypothesis are not clearly stated or declared.
- [4] The objectives are not explicitly stated and do not align with the conclusions.
- [5] The methodology is not clearly described, making it difficult to replicate the work.
- [6] The experimental setup and validation are not sufficiently detailed.
- [7] The results section lacks sufficient detail to support the stated objectives. Quantitative results are mentioned but not sufficiently elaborated or tied to figures/tables.
- [8] No baseline or state-of-the-art comparison is provided, which is critical for evaluating the novelty and effectiveness of the proposed workflow.
- [9] The Discussion section lacks a comprehensive comparison with prior work and does not interpret results in the context of existing literature.
- [10] The Conclusions section does not directly derive from the results and fails to answer the original hypothesis.
- [11] No clear statement of the study's limitations or contributions to the field.
- [12] The manuscript contains multiple instances of passive voice that obscure the clarity of the methodology.

## Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] [KEYWORDS] Keywords were not detected or fewer than three were found.
- [2] Section balance is uneven, with some sections disproportionately long or absent.
- [3] The references section is incomplete and lacks proper formatting.
- [4] The problem statement lacks specific objectives and challenges.
- [5] The model and method sections are too vague.
- [6] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.
- [7] The phrase 'workflow' is used ambiguously in the context of patient selection criteria.
- [8] The Discussion section is incomplete and lacks depth.
- [9] The Conclusions section is vague and not quantified.
- [10] Some technical terms are defined on first use, which is a positive aspect.
- [11] The abstract is cut off and incomplete, which affects the overall readability.
- [12] The manuscript could benefit from a more detailed description of the methodology and results.

## Specific Improvement Recommendations

1. [KEYWORDS] Provide 4-8 indexing terms after the abstract.
2. Add an introduction and literature review section to provide context and establish the research problem.
3. Reorganize the manuscript to follow the standard structure: Introduction → Literature → Methods → Results → Discussion → Conclusions.
4. Clearly state the research hypothesis and objectives in the introduction.
5. Ensure that the objectives are explicitly addressed in the conclusions.
6. Complete and properly format the references section.
7. Provide a detailed description of the workflow, including specific steps and tools used.

8. Include equations or mathematical formulations if applicable.
9. Clarify the experimental setup, dataset, and validation process.
10. Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
11. Include detailed quantitative metrics such as MSE, RMSE, CV, or hypervolume values with corresponding figures/tables.
12. Provide a clear comparison with existing methods or baselines to contextualize the results.

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